



Finding Open Data

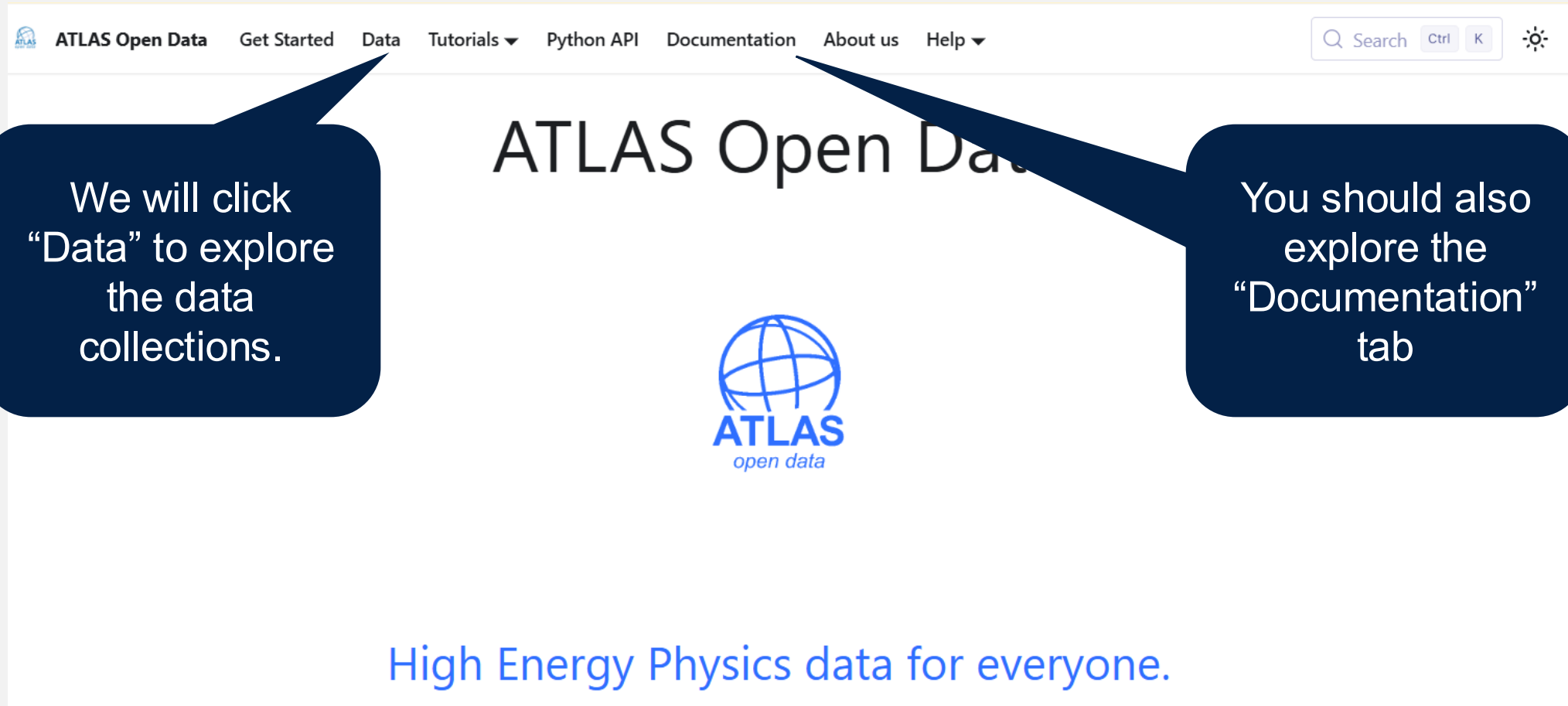
Finding Open Data

First, go to the ATLAS Open Data website:

<https://opendata.atlas.cern/>



Finding Open Data



The screenshot shows the ATLAS Open Data website. The navigation bar at the top includes links for ATLAS Open Data, Get Started, Data, Tutorials, Python API, Documentation, About us, and Help. A search bar is located on the right. The main heading is 'ATLAS Open Data' with the ATLAS open data logo below it. A blue tagline at the bottom reads 'High Energy Physics data for everyone.' Two dark blue callout boxes with white text provide instructions: one points to the 'Data' link in the navigation bar, and the other points to the 'Documentation' link.

ATLAS Open Data

Get Started Data Tutorials Python API Documentation About us Help

Search Ctrl K

ATLAS open data

High Energy Physics data for everyone.

We will click “Data” to explore the data collections.

You should also explore the “Documentation” tab

Finding Open Data

The Data

The ATLAS collaboration, dedicated to advancing scientific knowledge and promoting open science, has made a significant portion of its data publicly available. This release includes two primary types of data:

- [Monte Carlo \(MC\) Simulations](#): Simulated particle collisions within the ATLAS detector.
- [Detector Data](#): Measurements recorded from particle collisions within the ATLAS detector.

These datasets provide valuable insights into high-energy particle physics and are available in two levels of complexity: for educational purposes and for research. They are good resources for researchers, educators, and students, supporting a wide range of scientific inquiries, educational projects, and exploratory analyses.

For detailed information on the data content and how to download it, explore the categories below:



Accessing the data

3 items



Data for education

3 items

Read through
the information
linked here.

Then click on
Data for
Education

Finding Open Data

Data for education

For educational data, we currently offer two different datasets: one at a center-of-mass energy of 8 TeV, corresponding to data taken in 2012, and another at 13 TeV, corresponding to data taken in 2016.

 **13 TeV 2025 Data**

1 item

 **13 TeV 2020 Data**

2 items

 **8 TeV Data**

1 item

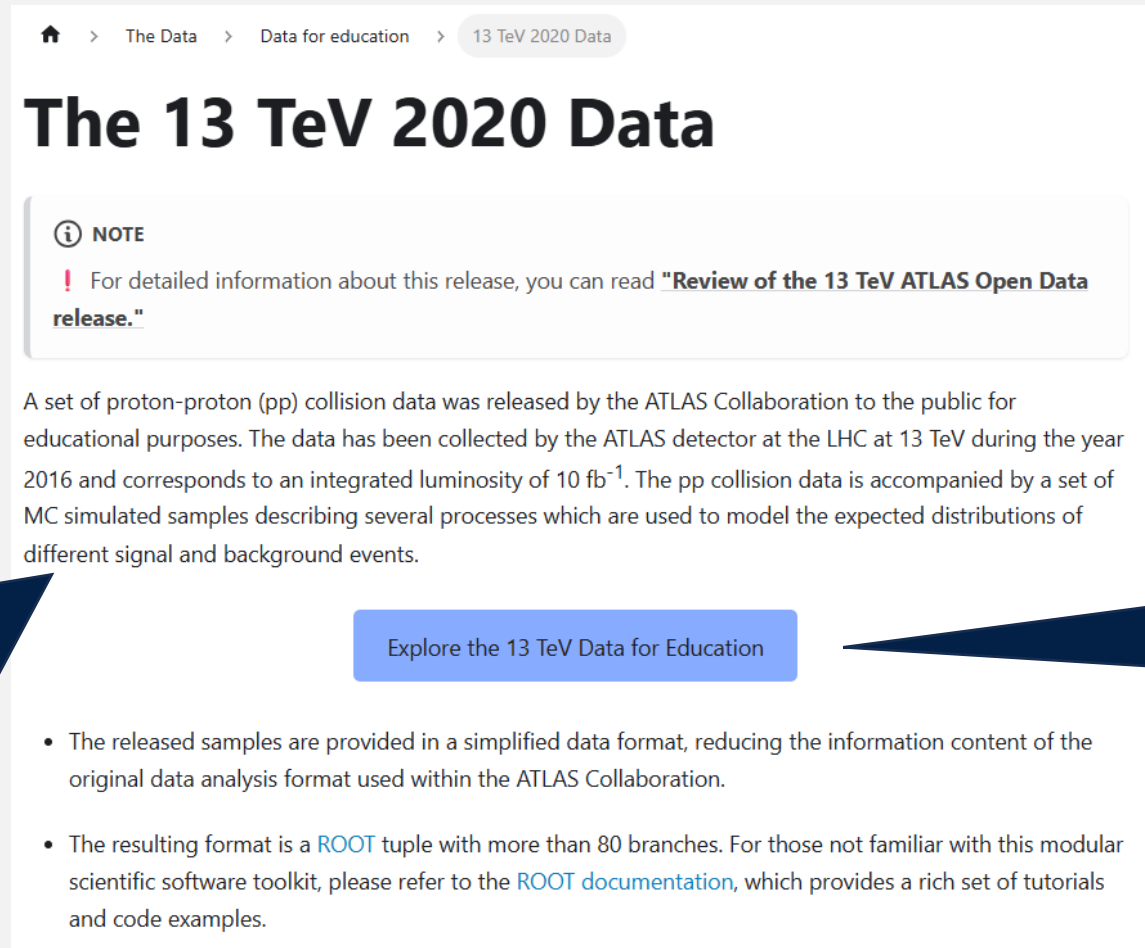
We will initially be using the 2020 data.

If you want to you can explore other data sets when conducting your own research in Phase 3

Finding Open Data

Read down through this page. Some parts might be confusing, but you will gradually develop familiarity with the notation and the contents of the files.

You will likely revisit it and the documentation tab a few times to read about the different types of analysis you can do and to find the numbers/names of datafiles for those analyses.



The screenshot shows a web page titled "The 13 TeV 2020 Data" with a breadcrumb trail: Home > The Data > Data for education > 13 TeV 2020 Data. Below the title is a "NOTE" section with an exclamation mark icon, stating: "For detailed information about this release, you can read 'Review of the 13 TeV ATLAS Open Data release.'" Below this is a paragraph of text: "A set of proton-proton (pp) collision data was released by the ATLAS Collaboration to the public for educational purposes. The data has been collected by the ATLAS detector at the LHC at 13 TeV during the year 2016 and corresponds to an integrated luminosity of 10 fb^{-1} . The pp collision data is accompanied by a set of MC simulated samples describing several processes which are used to model the expected distributions of different signal and background events." Below the text is a blue button labeled "Explore the 13 TeV Data for Education". At the bottom, there is a list of two bullet points: "The released samples are provided in a simplified data format, reducing the information content of the original data analysis format used within the ATLAS Collaboration." and "The resulting format is a ROOT tuple with more than 80 branches. For those not familiar with this modular scientific software toolkit, please refer to the ROOT documentation, which provides a rich set of tutorials and code examples."

Home > The Data > Data for education > 13 TeV 2020 Data

The 13 TeV 2020 Data

NOTE

! For detailed information about this release, you can read "[Review of the 13 TeV ATLAS Open Data release.](#)"

A set of proton-proton (pp) collision data was released by the ATLAS Collaboration to the public for educational purposes. The data has been collected by the ATLAS detector at the LHC at 13 TeV during the year 2016 and corresponds to an integrated luminosity of 10 fb^{-1} . The pp collision data is accompanied by a set of MC simulated samples describing several processes which are used to model the expected distributions of different signal and background events.

[Explore the 13 TeV Data for Education](#)

- The released samples are provided in a simplified data format, reducing the information content of the original data analysis format used within the ATLAS Collaboration.
- The resulting format is a [ROOT](#) tuple with more than 80 branches. For those not familiar with this modular scientific software toolkit, please refer to the [ROOT documentation](#), which provides a rich set of tutorials and code examples.

When you're ready, click the blue button to navigate to the files themselves.

Finding Open Data

You can use filters to narrow your search, e.g. if you want Monte Carlo simulated data you can select “Derived”

The screenshot shows the OpenData CERN search interface. The search bar contains the text "ATLAS 13 TeV for 2020 Open Data release". The results section shows 8 results found, sorted by "Best match". The left sidebar contains filters for "Current parameters", "Type", "Experiment", "Availability", and "Year". The "Type" filter is expanded, showing options like "Dataset (8)", "Derived (8)", "Simulated (8)", "Documentation (1)", "About (1)", "Environment (1)", "VM (1)", and "News (2)". The "Derived" option is selected. The "Experiment" filter shows "ATLAS (8)" selected. The "Availability" filter shows "online (8)" selected. The "Year" filter is empty. The search results list four entries, each with a title, description, and buttons for "Dataset", "Derived", "Simulated", and "ATLAS".

opendata
CERN

Help ▾ About ▾

ATLAS 13 TeV for 2020 Open Data release

Search

8 result(s) found

Sort by Best match

Current parameters Clear all

ATLAS X Dataset X

Type

- ☒ Dataset (8)
- ☐ Derived (8)
- ☐ Simulated (8)
- ☐ Documentation (1)
- ☐ About (1)
- ☐ Environment (1)
- ☐ VM (1)
- ☐ News (2)

Experiment

- ☒ ATLAS (8)

Availability

- ☐ online (8)

Year

ATLAS 13 TeV samples collection Gamma-Gamma, for 2020 Open Data release

The ATLAS Collaboration has released 10/fb of collision data at a centre-of-mass energy of 13 TeV from the 2016 data-taking period, intended solely for educational purposes, to the general public. ...

Dataset Derived Simulated ATLAS

ATLAS 13 TeV samples collection exactly two leptons (electron or muon), for 2020 Open Data release

The ATLAS Collaboration has released 10/fb of collision data at a centre-of-mass energy of 13 TeV from the 2016 data-taking period, intended solely for educational purposes, to the general public. ...

Dataset Derived Simulated ATLAS

ATLAS 13 TeV samples collection exactly three leptons (electron or muon), for 2020 Open Data release

The ATLAS Collaboration has released 10/fb of collision data at a centre-of-mass energy of 13 TeV from the 2016 data-taking period, intended solely for educational purposes, to the general public. ...

Dataset Derived Simulated ATLAS

ATLAS 13 TeV samples collection exactly one lepton (electron or muon), for 2020 Open Data release

The ATLAS Collaboration has released 10/fb of collision data at a centre-of-mass energy of 13 TeV from the 2016 data-taking period, intended solely for educational purposes, to the general public. ...

Dataset Derived Simulated ATLAS

ATLAS 13 TeV samples collection at least four leptons (electron or muon), for 2020 Open Data release

The ATLAS Collaboration has released 10/fb of collision data at a centre-of-mass energy of 13 TeV from the 2016 data-taking period, intended solely for educational purposes, to the general public. ...

Dataset Derived Simulated ATLAS

Then you can click on a row to look at it in detail.

Finding Open Data

ATLAS 13 TeV samples collection exactly two leptons (electron or muon)

ATLAS Collaboration

Cite as: ATLAS Collaboration (2020). ATLAS 13 TeV samples collection exactly two leptons (electron or muon).
DOI: [10.7483/OPENDATA.ATLAS.OHZ7.RFNH](https://doi.org/10.7483/OPENDATA.ATLAS.OHZ7.RFNH)

Dataset Derived Simulated ATLAS 13TeV pp CERN-LHC

Description

The ATLAS Collaboration has released 10/fb of collision data at a centre-of-mass energy of 13 TeV from the 2016 data-taking period, intended solely for educational purposes, to the general public. This set of real data is accompanied by matching simulated samples of Standard Model processes and a selection of Beyond the Standard Model signals.

Dataset characteristics

210 files. **47.7 GiB** in total.

How were these data generated?

Events selected with exactly two leptons (electron or muon). Both real data and simulated samples are subjected to a loose event preselection to reduce the processing time by reducing the overall number of events that have to be analysed. The preselection consists of a set of object selection criteria and a subsequent event selection acting on these preselected objects.

These DOI links are unique identifiers that can be used with the opendata client and the FileRetriever.py “print_files” method.

Finding Open Data

This dataset is provided by the ATLAS Collaboration **only for educational purposes and is not suited for scientific publications**. You can access through the ATLAS Virtual Machine and Software. More documentation is available at:

[The ATLAS Open Data documentation for 13 TeV release](#)

[The ATLAS Open Data site](#)

Files and indexes

Filename	Size	
data_A.exactly2lep.root	122.1 MiB	Download
data_B.exactly2lep.root	450.8 MiB	Download
data_C.exactly2lep.root	657.0 MiB	Download
data_D.exactly2lep.root	1009.4 MiB	Download
exactly2lep.zip	20.9 GiB	Download

[«](#) [<](#) [1](#) [2](#) [3](#) [4](#) [5](#) [...](#) [42](#) [>](#) [»](#)

There are links to the documentation

Sometimes there are a lot of tiles!

It is possible to download the files and work with them locally, but as they are very large this is inefficient. It also doesn't work well with the Monty server.

Finding Open Data

ATLAS 13 TeV samples collection exactly two leptons (electron or muon)

ATLAS Collaboration

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DOI: [10.7483/OPENDATA.ATLAS.OHZ7.RFNH](https://doi.org/10.7483/OPENDATA.ATLAS.OHZ7.RFNH)

[Dataset](#) [Derived](#) [Simulated](#) [ATLAS](#) [13TeV](#) [pp](#) [CERN-LHC](#)

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So instead, copy and paste this DOI and bring it back to your notebook.

Finding Open Data

```
#Import the uproot library
import uproot

#Here you could also import any other python libraries you would like to use
import matplotlib.pyplot as plt
import numpy as np

#Function to list datafiles from an Open Data collection
from FileRetriever import print_files as print_files
```

In Notebook 3 we show you how to import the `print_files` method from the `FileRetriever`

```
#An example of how to print a list of the urls for e.g. https://opendata.cern.ch/record/15005
print_files("10.7483/OPENDATA.ATLAS.2Y1T.TLGL") #Parameter = DOI from the top of the page, as a string
```

This is how to use the DOI from the top of the page to retrieve the list of urls for the data files.

```
http://opendata.cern.ch/eos/opendata/atlas/OutreachDatasets/2020-08-19/4lep.zip
http://opendata.cern.ch/eos/opendata/atlas/OutreachDatasets/2020-08-19/4lep/Data/data_A.4lep.root
http://opendata.cern.ch/eos/opendata/atlas/OutreachDatasets/2020-08-19/4lep/Data/data_B.4lep.root
http://opendata.cern.ch/eos/opendata/atlas/OutreachDatasets/2020-08-19/4lep/Data/data_C.4lep.root
http://opendata.cern.ch/eos/opendata/atlas/OutreachDatasets/2020-08-19/4lep/Data/data_D.4lep.root
http://opendata.cern.ch/eos/opendata/atlas/OutreachDatasets/2020-08-19/4lep/MC/mc_301215.ZPrime2000_ee.4lep.root
http://opendata.cern.ch/eos/opendata/atlas/OutreachDatasets/2020-08-19/4lep/MC/mc_301216.ZPrime3000_ee.4lep.root
http://opendata.cern.ch/eos/opendata/atlas/OutreachDatasets/2020-08-19/4lep/MC/mc_301220.ZPrime2000_mumu.4lep.root
http://opendata.cern.ch/eos/opendata/atlas/OutreachDatasets/2020-08-19/4lep/MC/mc_301221.ZPrime3000_mumu.4lep.root
http://opendata.cern.ch/eos/opendata/atlas/OutreachDatasets/2020-08-19/4lep/MC/mc_301322.ZPrime400_tt.4lep.root
http://opendata.cern.ch/eos/opendata/atlas/OutreachDatasets/2020-08-19/4lep/MC/mc_301323.ZPrime500_tt.4lep.root
http://opendata.cern.ch/eos/opendata/atlas/OutreachDatasets/2020-08-19/4lep/MC/mc_303329.RS_G_ZZ_1111_c10_m1000.4lep.root
http://opendata.cern.ch/eos/opendata/atlas/OutreachDatasets/2020-08-19/4lep/MC/mc_303334.RS_G_ZZ_1111_c10_m2000.4lep.root
http://opendata.cern.ch/eos/opendata/atlas/OutreachDatasets/2020-08-19/4lep/MC/mc_305553.Gee_01_1000.4lep.root
http://opendata.cern.ch/eos/opendata/atlas/OutreachDatasets/2020-08-19/4lep/MC/mc_305556.Gee_01_2000.4lep.root
```

Copy the link of the file that you want.

Finding Open Data

```
#f = uproot.open("")  
f = uproot.open("http://opendata.cern.ch/eos/opendata/atlas/OutreachDatasets/2020-08-19/11largeRjet1lep/Data/data\_A.11largeRjet1lep.root")
```

Then paste it into the uproot.open method as a string (i.e. between two double quotes “ ”)

Congratulations! You've successfully loaded a data file into your Jupyter Notebook!